Currently, no purely subtractive system is known (in which addition does not play some role), and all subtractive systems use both addition and subtraction. Most subtractive systems only employ subtraction for 1 (therefore, for example, in such a base-10 subtractive system, the number 38 is written as $30 + 8$, but the number 39 is written as $40 - 1$).

## Problem 9.6

### Yoruba (Harold Somers, UKLO 2020)

Here are some numbers in Yoruba:

| | | | |
|---|---|---|---|
| *èji* | 2 | *ẹ́ẹ́rìndilogóji* | 36 |
| *ẹ̀rin* | 4 | *ẹ̀rìndogóji* | 44 |
| *àrun* | 5 | *àádorin* | 70 |
| *ẹ̀rinlá* | 14 | *ẹ́ẹ́tàdilogórin* | 77 |
| *eéjìdilogun* | 18 | *ẹ̀tàdogórin* | 83 |

**Problem 9.6a** Write in numerals:

a. *àádota*
b. *àrùndogórin*
c. *aárùndilogórin*
d. *ẹ̀tàdogórun*
e. *òkándilogóji*

**Problem 9.6b** Write in Yoruba: 12, 45, 57, 90, 99.



The marks above the vowels denote tones; *e* and *ẹ* are distinct vowels.

### Solution

Firstly, we can notice the pair 4 and 14, in which the only difference is the suffix *-lá*. We can therefore deduce that $10 + X = X$*-lá*, so this system is most likely base-10.

Next, we have the pair 36 and 44, in which the only difference is *-il-*. Moreover, the first part of the word is highly similar to the word for 4 (in the case of 44 it is actually identical, while in the case of 36 it is slightly mutated), and the